

Session Timings

9.30AM – 10.45AM –P1

15min Break

11.00AM to 1.00PM – P2

1 Hr Lunch break

2.00PM to 3.45PM –P3

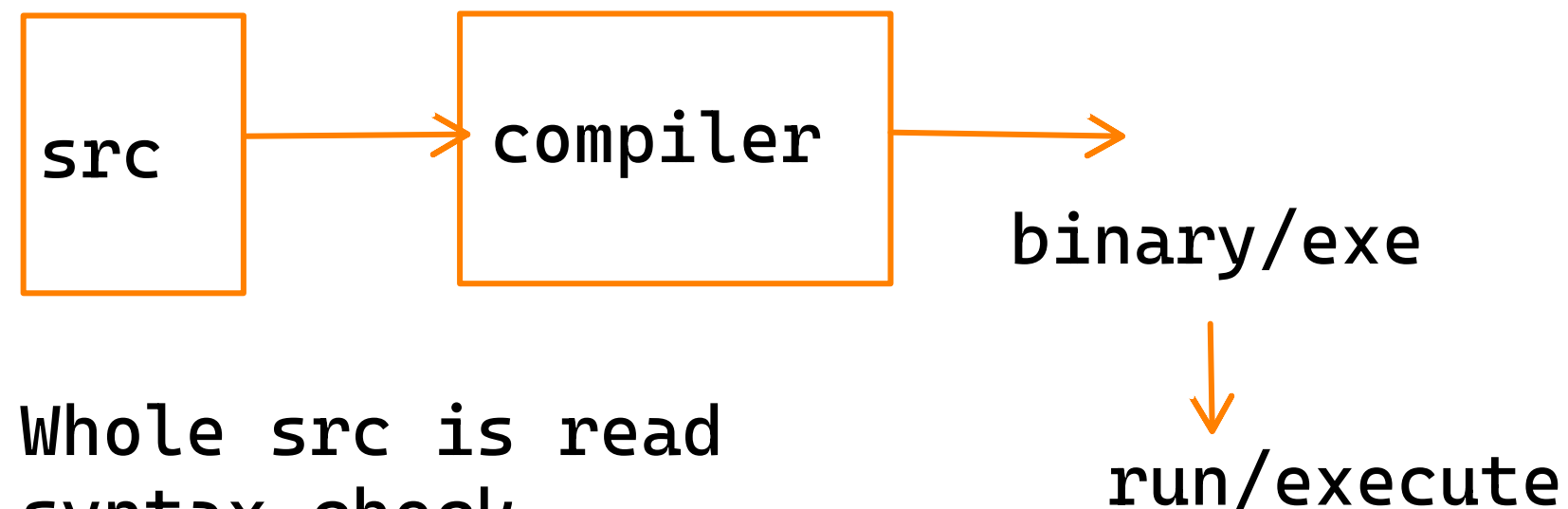
15min Break

4.00PM to 5.30PM – P4

Computers work on Numbers - 1/0 - binary

Any Programming Language → High Level → Binary

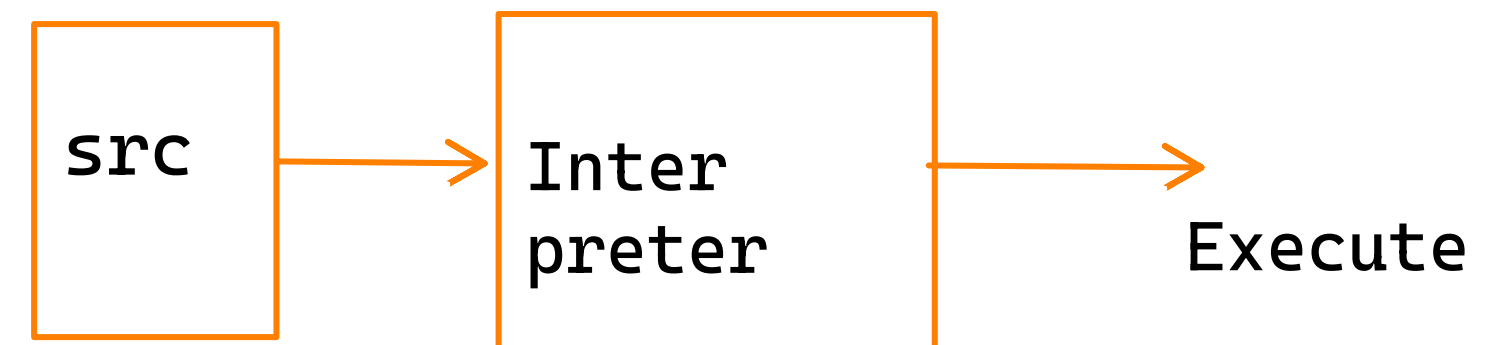
Compilation



Whole src is read
syntax check
linking
convert to binary

Eg: C, C++, Go, Rust..

Interpretation



Reading command by command → convert to bin
→ Execute the command

Eg: Python, JS

Python Programming Language

1. Interpreted
2. Open Source
3. Dynamic
4. Platform Independent
5. WORA – Write Once Run Anywhere
6. Huge Libraries

Versions of Python

2.x.x  3.x.x

3.9.10  3.12.12  3.14.14

Things needed

Interpreter

Text Editor – VSCode

Interpreters of Python

Cython – C Implementation
Jython – Java Implementation
Rpython – R Implementation
Pypi – Python Implementation

Working with Python

1. Interactive Mode

- No Saving of commands
- Use Cases:
 - Prototyping
 - Help

```
C:\Users\VishwasKSingh>python
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> print("Hello World")
Hello World
>>> 2+3
5
>>> 45*2
90
>>> dir(str)
['__add__', '__class__', '__contains__', '__delattr__', '__dir__', '__doc__', '__eq__', '__format__', '__ge__', '__getattribute__', '__getitem__', '__getnew
args__', '__getstate__', '__gt__', '__hash__', '__init__', '__init_subclass__', '__iter__', '__le__', '__len__', '__lt__', '__mod__', '__mul__', '__ne__', '
__new__', '__reduce__', '__reduce_ex__', '__repr__', '__rmod__', '__rmul__', '__setattr__', '__sizeof__', '__str__', '__subclasshook__', 'capitalize', 'case
fold', 'center', 'count', 'encode', 'endswith', 'expandtabs', 'find', 'format', 'format_map', 'index', 'isalnum', 'isalpha', 'isascii', 'isdecimal', 'isdigi
t', 'isidentifier', 'islower', 'isnumeric', 'isprintable', 'isspace', 'istitle', 'isupper', 'join', 'ljust', 'lower', 'lstrip', 'maketrans', 'partition', 'r
emoveprefix', 'removesuffix', 'replace', 'rfind', 'rindex', 'rjust', 'rpartition', 'rsplit', 'rstrip', 'split', 'splitlines', 'startswith', 'strip', 'swapca
se', 'title', 'translate', 'upper', 'zfill']
>>> help(str.index)
Help on method_descriptor:

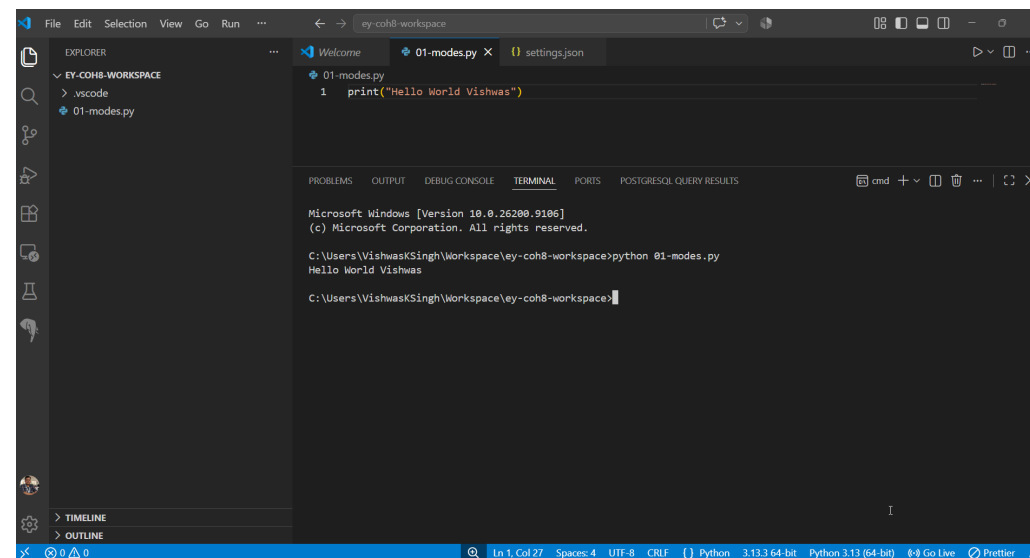
index(self, sub[, start[, end]], /) unbound builtins.str method
    Return the lowest index in S where substring sub is found, such that sub is contained within S[start:end].

    Optional arguments start and end are interpreted as in slice notation.
    Raises ValueError when the substring is not found.

>>> quit()
```

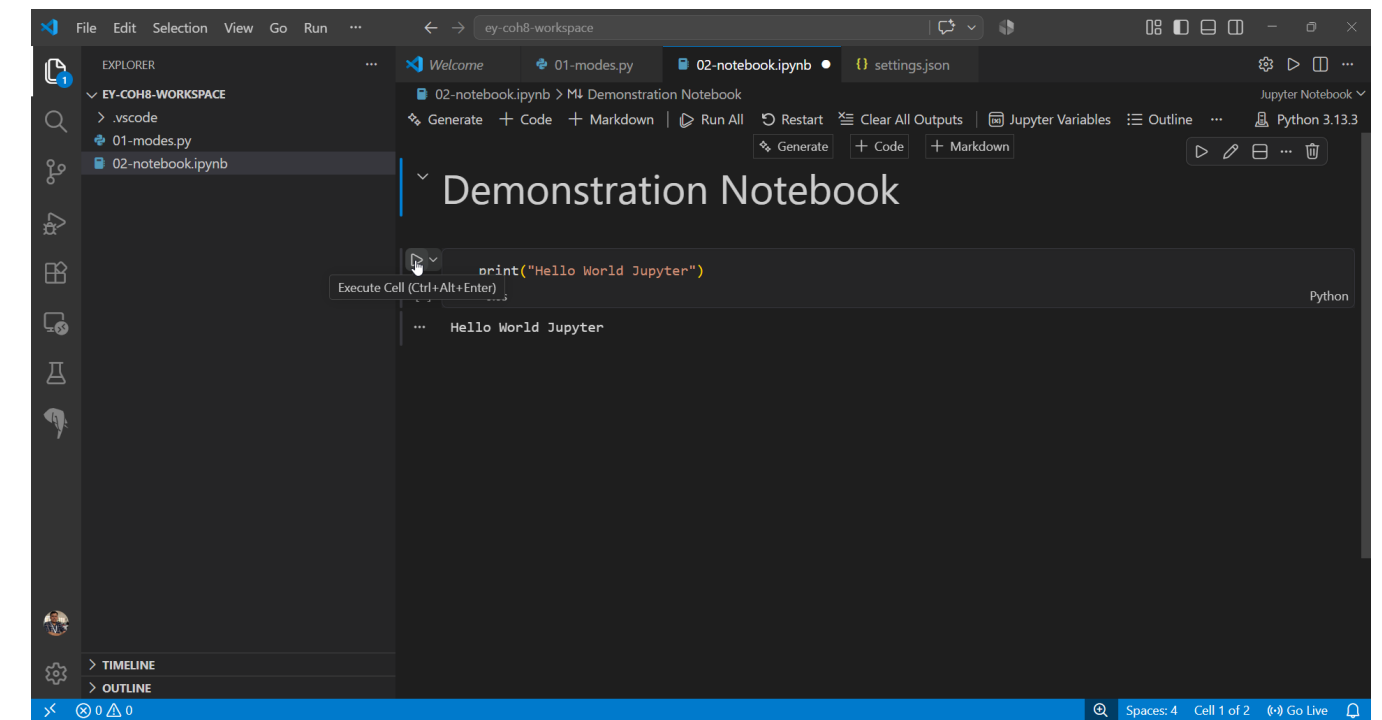
2. Script Mode

- Re-run codes
- run in backends
- Use Case:
 - Batch Processing
 - Automating
 - many more ...



3. Notebook Mode

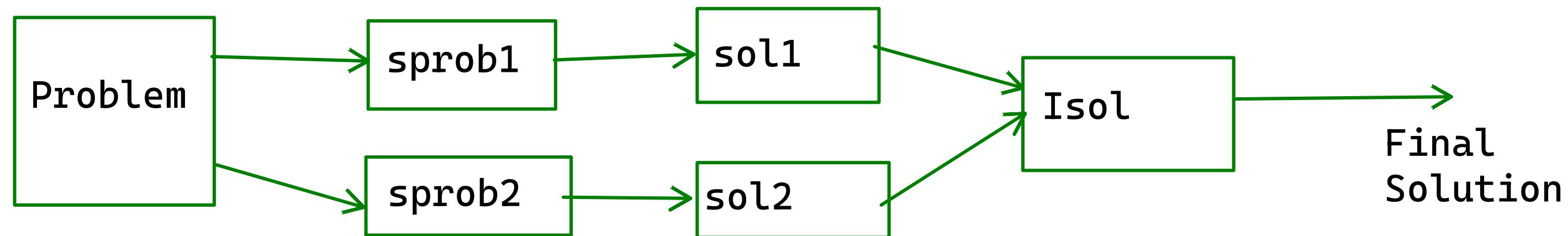
- Jupyter Notebook/Interactive Python
- Google Colab
- Use Case:
 - Analyst, Data Scientist



Programming Language

1. Base Paradigm
 - Functional Programming
2. How store data temporarily
3. How to branch using conditions
4. How to repeat

Functional Programming Approach



Eg: You are an Analyst, you get data from various depts(Excel, csv, pdf..). Your job is to give consolidated report

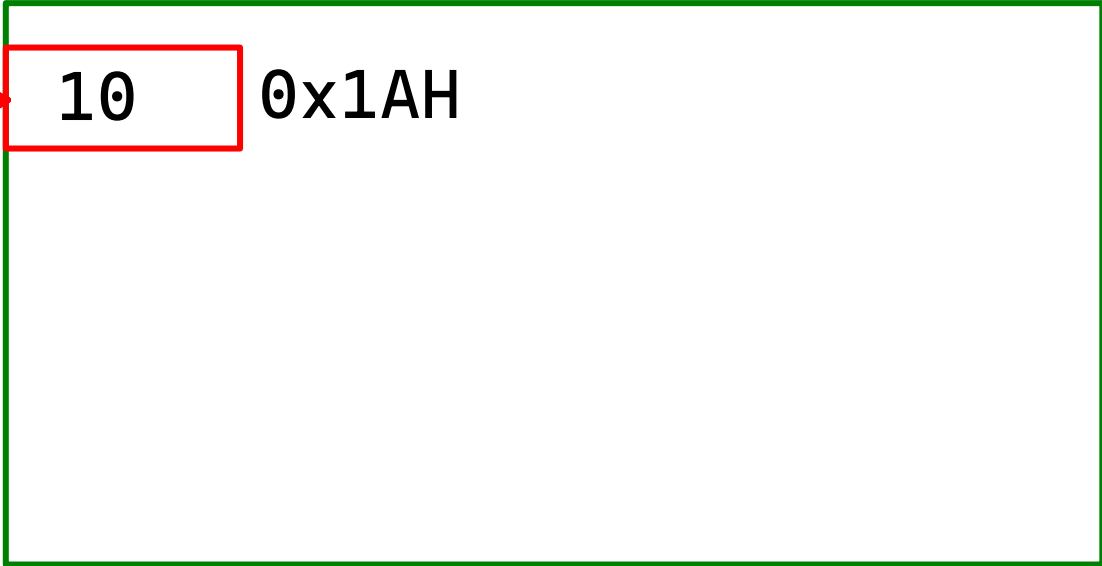
1. reading data sources
2. cleaning the sources
3. process it
4. Generate a report from consolidated data

- read_pdf(), read_excel() ...
- handle_missing_values(), fix_formats()
- analyse_data(), ...
- generate_pdf(), generate_html()....

step-by-step
call

Variables

Label to Memory Location

age  

 age = 10
variable assgn data
identifier op

Dynamic Programming Language

Not specifying Data type when we define variables

Rules

1. Can contain only Numbers/letters/underscore(_)
2. Cannot start with a number
3. It cannot be a keyword
4. Case Sensitive

```
Type "help", "copyright", "credits" or
>>> x = 10
>>> x1 = 100
>>> 1x
      File "<python-input-2>", line 1
        1x
        ^
SyntaxError: invalid decimal literal
>>> x_1 = 123
>>> X1 = 45.24
>>> |
```

```
>>> import keyword
>>> keyword.kwlist
['False', 'None', 'True', 'and', 'as', 'assert', 'async', 'await', 'break', 'class', 'continue', 'def', 'del', 'elif', 'else', 'except', 'finally', 'for', 'from', 'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not', 'or', 'pass', 'raise', 'return', 'try', 'while', 'with', 'yield']
>>>
```

1. Use meaningful words
2. We use snake_case, Eg: no_of_bottles, price_of_item

Data Types

```
>>> name = "vishwas"
>>> type(name)
<class 'str'>
>>> age = 10
>>> type(age)
<class 'int'>
>>> price = 23.45
>>> type(price)
<class 'float'>
>>> is_active = True
>>> type(is_active)
<class 'bool'>
>>> complex_number = complex(2,4)
>>> type(complex_number)
<class 'complex'>
>>> ip_addresses = ['192.168.1.1', '23.45.67.9']
>>> type(ip_addresses)
<class 'list'>
>>> inventory = {'apple': 25, 'mango': 27}
>>> type(inventory)
<class 'dict'>
>>> version = (12, 1, 1)
>>> type(version)
<class 'tuple'>
>>> distinct_vals = {1, 2, 34}
>>> type(distinct_vals)
<class 'set'>
>>> |
```

```
>>> x = 1234567987729018291820810289172918102819172918280301818291891821980197918331080189182
>>> x
1234567987729018291820810289172918102819172918280301818291891821980197918331080189182
>>> x1 = 10_00_000
>>> x1
1000000
>>> x2 = 0b10101
>>> x2
21
>>> type(x2)
<class 'int'>
>>> x3 = 0x1234
>>> x3
4660
>>> type(x3)
<class 'int'>
>>> x4 = 123e10
>>> x4
12300000000000.0
>>> x5 = -10
>>> x5
-10
```

Text Representation in python as strings

```
>>> s1 = 'vishwas'
>>> s2 = "vishwas"
>>> s3 = '''vishwas'''
>>> type(s3)
<class 'str'>
>>> type(s2)
<class 'str'>
>>> type(s1)
<class 'str'>
>>> s4 = "This is Vishwas's Book"
>>> s5 = 'He said "He is Learning Python"'
>>> s6 = '''This is Vishwas K Singh.
... I am Learning Python.
... Today is 21 Sept 2026'''
>>> s6 = """This is Vishwas K Singh.
... I am Learning Python.
... Today is 21 Sept 2026"""
>>> s7 = 'C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace\01-modes.py'
File "<python-input-49>", line 1
    s7 = 'C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace\01-modes.py'
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
SyntaxError: (unicode error) 'unicodeescape' codec can't decode bytes in posi
>>> s7 = r'C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace\01-modes.py'
>>> s7
'C:\\Users\\VishwasKSingh\\Workspace\\ey-coh8-workspace\\01-modes.py'
>>> s8 = f"This is {s1}"
>>> s8
'This is vishwas'
>>>
```

Output / Displaying Values

```
>>> s6
'This is Vishwas K Singh.\nI am Learning Python.\nToday is 21 Sept 2026'
>>> s7
'C:\\Users\\VishwasKSingh\\Workspace\\ey-coh8-workspace\\01-modes.py'
>>>
```

```
01-modes.py > ...
1  print("Hello World Vishwas")
2  age = 10
3  age
4  print(age)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER POSTGRESQL QUERY RESULTS

```
C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace>python 01-modes.py
Hello World Vishwas
10

C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace>
```

```
>>> print(s6)
This is Vishwas K Singh.
I am Learning Python.
Today is 21 Sept 2026
>>> print(s7)
C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace\01-modes.py
>>> |
```

```
01-modes.py > ...
1  print("Hello World Vishwas")
2  age = 10
3  # This is a comment, not processed by interpreter
4  # age
5  print(age)
6  print("I am", age, "years old") # This is a print statement, combining multiple strings
7
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER POSTGRESQL QUERY RESULTS

10

```
C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace>
C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace>python 01-modes.py
Hello World Vishwas
10
I am 10 years old
```

Various ways of Print Statements

01-modes.py > ...

```
1  print("Hello World Vishwas")
2  age = 10
3  language = "python"
4  version = 3.14
5  print("I am learning", language, version)
6  print("I am learning "+language+" "+str(version))
7  print("I am learning %s %f"%(language,version))
8  print("I am learning %s %.2f"%(language,version))
9  print("I am learning {0} {1}".format(language,version))
10 # Most used way - RECOMMENDED
11 print(f"I am learning {language} {version:.2f}")
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

JUPYTER

POSTGRESQL QUERY RESULTS

C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace>python 01-modes.py

Hello World Vishwas

I am learning python 3.14

I am learning python 3.14

I am learning python 3.140000

I am learning python 3.14

I am learning python 3.14

I am learning python 3.14

C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace>

03-simple_interest.py > ...

```
1 principal = 10_00_00
2 time_in_months = 36
3 rate_of_interest = 0.07
4
5 simple_interest = (principal*time_in_months*rate_of_interest)/100
6
7 print(f"The Amount Rs.{principal} kept in our bank for {time_in_months} months, gives Rs.{s
```

I

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER POSTGRESQL QUERY RESULTS

```
C:\Users\VishwasKSingh\Workspace\ey-coh8-workspace>python 03-simple_interest.py
The Amount Rs.100000 kept in our bank for 36 months, gives Rs.2520.00 at 7.00% rate of interest
```

Lab: Brokerage Calculator

Brokerage Calculator

Equity delivery	
Brokerage	Zero Brokerage
STT/CTT	0.1% on buy & sell
Transaction charges	NSE: 0.00307% BSE: 0.00375%
GST	18% on (brokerage + SEBI charges + transaction charges)
SEBI charges	₹10 / crore
Stamp charges	0.015% or ₹1500 / crore on buy side

Making our scripts dynamic

→ `v1 = input("Enter you Name: ")`

```
>>> v1 = input("Enter your name")
Enter your namevishwas
>>> v1 = input("Enter your name: ")
Enter your name: vishwas
>>> v2 = input("Enter the price of onion: ")
Enter the price of onion: 50.50
>>> v2
'50.50'
>>> type(v2)
<class 'str'>
>>> |
```

Input taken from the user through console is always a string

```
>>> no_of_flags = int(input("Enter the number of flags: "))
Enter the number of flags: 30
>>> type(no_of_flags)
<class 'int'>
>>> no_of_flags
30
>>> |
```

Converting data types

```
>>> v2
'50.50'
>>> type(v2)
<class 'str'>
>>> v3 = float(v2)
>>> v3
50.5
>>> type(v3)
<class 'float'>
>>> |
```

```
>>> age = '10'
>>> type(age)
<class 'str'>
>>> age = int(age)
>>> age
10
>>> type(age)
<class 'int'>
>>> |
```

Operators in Python

```
>>> 2 > 3
False
>>> 4 > 5
False
>>> 3 < 8
True
>>> 2 > 3 and 4 < 5
False
>>> 8 > 9 or 2 < 3
True
>>> 'a' in 'cat'
True
>>> 'x' not in 'cat'
True
>>> v1 = "vishwas"
>>> v2 = "vishwas"
>>> v1 is v2
True
>>> v1 == v2
True
```

Internals of python

```
>>> v7 = 254
>>> v8 = 254
>>> id(v7)
140712363770696
>>> id(v8)
140712363770696
>>> v9 = 260
>>> v10 = 260
>>> id(v9)
3160909780496
>>> id(v10)
3160909780240
>>> v9 is v10
False
>>> v7 is v8
True
>>> v9
260
>>> del v9
>>>
```

- Any number non zero is considered True +ve & -ve both
- 0 is always false

In an AND statement

the second expr is evaluated only when 1st expr is True

In a OR statement

if the 1st expr is True, 2nd expr is not evaluated at all

Short circuit evaluation

Guard Pattern

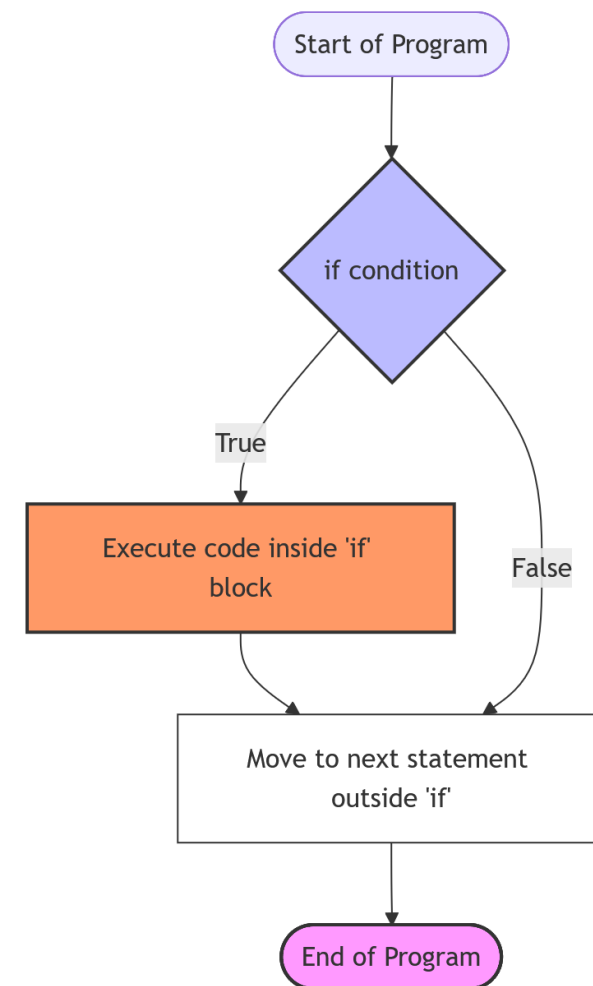
```
>>> x = 10
>>> y = 2
>>> x / y
5.0
>>> y = 0
>>> x / y
Traceback (most recent call last):
  File "<python-input-130>", line 1, in <module>
    x / y
    ~^~
ZeroDivisionError: division by zero
>>> y != 0 and x / y
False
>>> y = 2
>>> y != 0 and x / y
5.0
>>> |
```

```
>>> num = input("Enter a Number: ")
Enter a Number: a
>>> num
'a'
>>> num.isdigit() and int(num)
False
>>> num = input("Enter a Number: ")
Enter a Number: 123
>>> num.isdigit() and int(num)
123
>>>
```

```
>>> 0 and 4
0
>>> 4 and 0
0
>>> 2 and 4
4
>>> 4 and 2
2
>>> 2 and -3
-3
>>> -3 and 6
-3
>>> 0 or 2
2
>>> 2 or 0
2
>>> 5 or 4
5
>>> 6 or 3
6
>>>
```

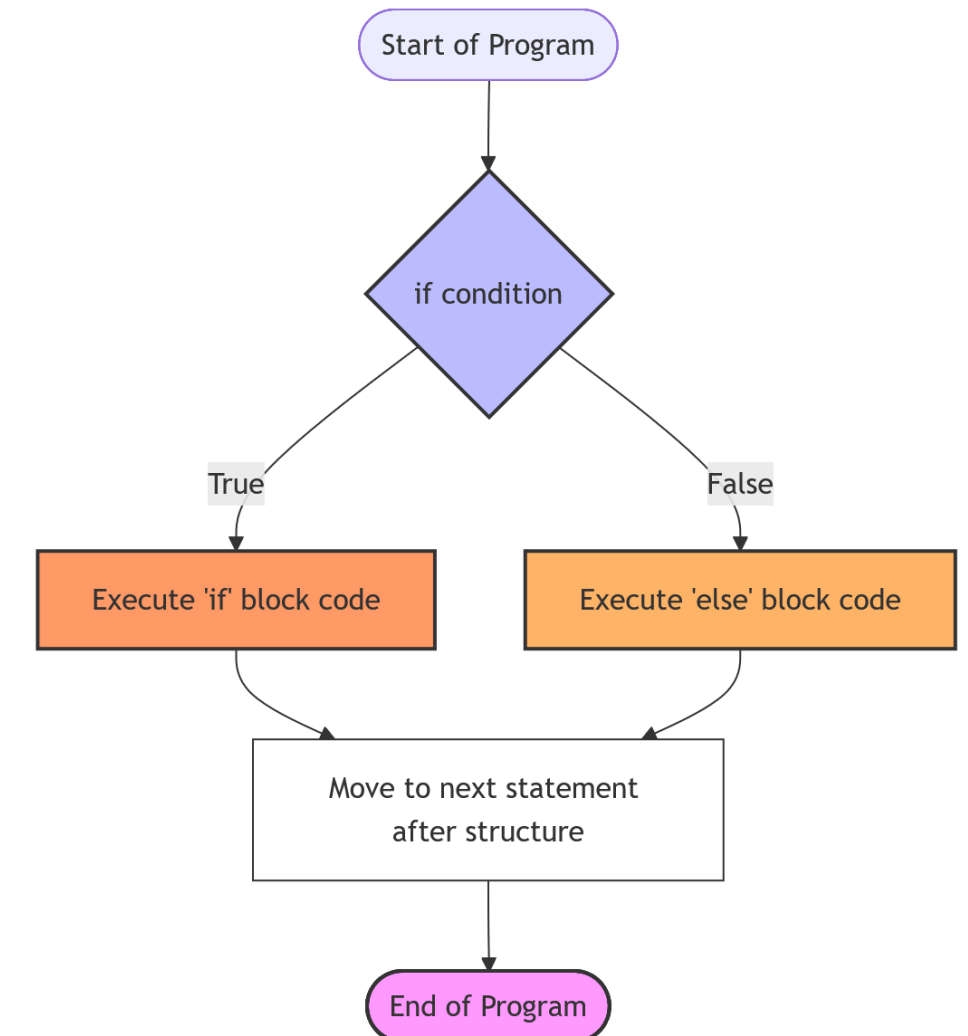
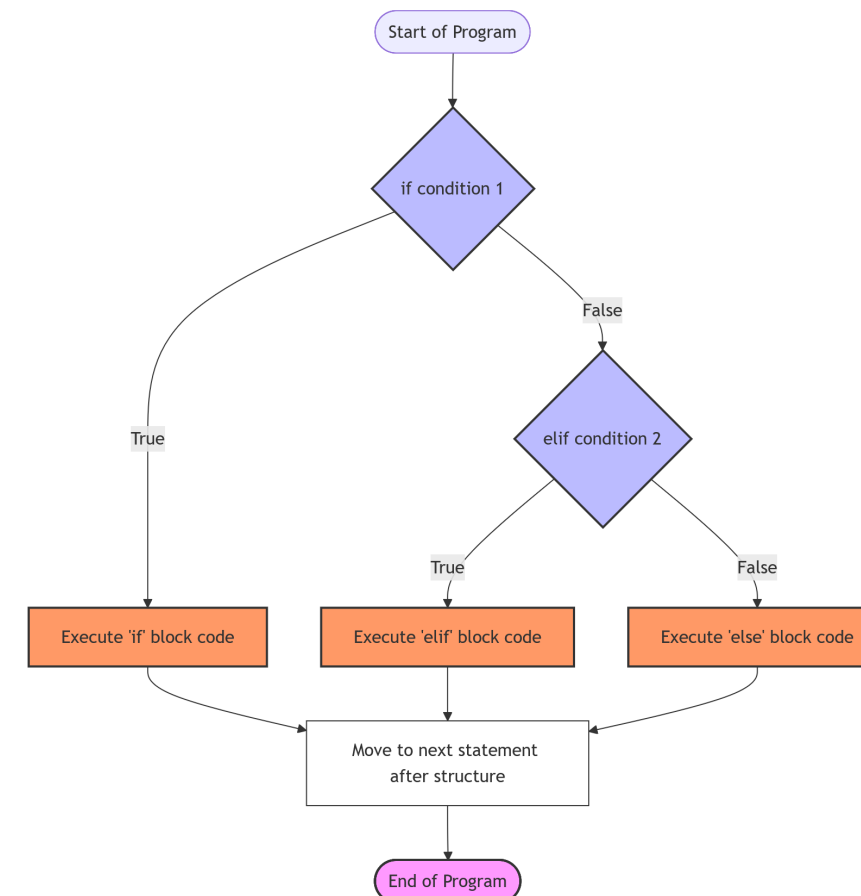
Branching code in python

```
if condn:  
    statement1  
statement2
```



```
if condn:  
    statement1  
else:  
    statement2  
statementn
```

if-elif-else



Lab: Build an income tax calculator

You take

1. age
2. income

based on that calculate the tax. Assume we calculate for new regime.

Loops/ Repetition

```
for i in range(10):  
    print(i)
```

```
class range(object)  
| range(stop) -> range object  
| range(start, stop[, step]) -> range object  
  
| Return an object that produces a sequence of integers from start (inclusive)  
| to stop (exclusive) by step. range(i, j) produces i, i+1, i+2, ..., j-1.  
| start defaults to 0, and stop is omitted! range(4) produces 0, 1, 2, 3.  
| These are exactly the valid indices for a list of 4 elements.  
| When step is given, it specifies the increment (or decrement).
```

```
>>> for i in range(5):  
...     print(i)  
...  
0  
1  
2  
3  
4
```

```
>>> for i in range(25,0,-1):  
...     print(i)  
...  
25  
24  
23  
22  
21  
20  
19  
18  
17  
16  
15  
14  
13  
12  
11  
10  
9  
8  
7  
6  
5  
4  
3  
2  
1
```

while

```
>>> x = 1
>>> while x <= 10:
...     print(x)
...     x += 1
...
1
2
3
4
5
6
7
8
9
10
>>>
```

```
>>> x = 0
>>> while(True):
...     print(x)
...     x+=10
...     if x == 100:
...         break
...
0
10
20
30
40
50
60
70
80
90
>>>
```

Lab: Take n numbers from the user, until he/she enters -999.
Find the total and average of n numbers excluding -999

Which loop will you choose?

1. You are building a log analyzer which contains 1000 records of logs.
2. You are building a ip streaming generator, where you want to stream ips randomly until the user stops it.

PEP8 - Style Guide

Linter - Readability

pylint, flake8, pep8....

Summary

- Python Programming Language basics
- Variables, data types, operators, if, Loops
- Style guide - PEP8